III. LESSON PROPER

1. **Routine: Opening and Closing Prayers**

Greetings! Magandang Buhay w/ Magalang Bow
Quotes/Bible Verse

Checking of Attendance

TMA (Teacher Made-Activity) - Engagement Activity (EA) - Objects of My Eye!

2. **Class Sharing**

Ask the students about Why it is important to measure usability goals?

Usability Goals and Metrics

Students will Appreciate the diverse components of UX and their impact on overall user satisfaction

3. **Class Discussion/Lecture**

Discussion of the Lesson/s in Week 3

1. Usability Goals and Metrics

Usability Goals

- Effectiveness: The accuracy and completeness with which users achieve their goals.
- Efficiency: The resources expended in relation to the accuracy and completeness of goals achieved. This includes the time taken to complete tasks.
- Satisfaction: The comfort and acceptability of the work system to its users and other stakeholders. This involves user attitudes and perceptions.

Usability Metrics

- Task Success Rate: The percentage of correctly completed tasks.
- Time on Task: The amount of time users take to complete a task.
- Error Rate: The frequency of errors made by users while performing tasks.
- Learnability: How quickly users can learn to use a system.
- Retention Rate: How well users remember how to use a system after a period of non-use.
- User Satisfaction: Often measured through surveys or questionnaires, such as the System Usability Scale (SUS).

Design Implications

- Interfaces should be designed to maximize effectiveness, efficiency, and satisfaction.
- Usability testing should be conducted to identify and address issues related to these goals.

2. User Experience (UX) Components

UX Components

- Usability: As described above, it includes effectiveness, efficiency, and satisfaction.
- Accessibility: Ensuring the system is usable by people with a wide range of abilities and disabilities.
- Desirability: The extent to which the user finds the product aesthetically pleasing

and enjoyable to use.

- Value: The extent to which the product fulfills the user’s needs and provides value.
- Findability: The ease with which users can find the information they need.
- Credibility: The degree to which the user perceives the product as trustworthy and reliable.

Holistic UX

- A good user experience encompasses all these components, ensuring that users find the product not only usable but also valuable, desirable, and accessible.

3. Measuring UX

Methods for Measuring UX

- Usability Testing: Observing users as they interact with the system to identify usability issues.
- Surveys and Questionnaires: Collecting subjective data from users about their experiences, such as SUS, Net Promoter Score (NPS), and custom surveys.
- Analytics and Metrics: Using tools to track user behavior, such as time on task, click paths, and error rates.
- A/B Testing: Comparing two versions of an interface to see which one performs better in terms of user experience.
- Heuristic Evaluation: Experts reviewing the interface against established usability principles (heuristics).

Applying UX Measurements

- Identifying Pain Points: Use data from usability testing and surveys to identify and prioritize areas for improvement.
- Iterative Design: Continuously improve the interface based on user feedback and testing results.
- User-Centered Design: Involve users throughout the design process to ensure the final product meets their needs and expectations.

Measuring the Intangible. Usability Metrics

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MIN TO READ

How do we evaluate the design? Typically, the first ones to approve it are the designers themselves. After that, other decision-makers give their opinion. Then, some users test it and give their feedback.

Having a couple of levels of approval is great, but all those opinions are rather subjective. As a design agency, we always promote a data-driven approach. And in the case of measuring user experience, usability metrics are *essential data*.

Need a data-driven approach to UX?

What are usability metrics?

Usability metrics are a system of measurement of the effectiveness, efficiency, and satisfaction of users working with a product.

To put it simply, such metrics are used to measure how easy and effective the product is for users.

Most usability metrics are calculated based on the data collected during usability testing. Users are asked to complete a **task** while researchers observe the user behavior and take notes. A task can be "Find the price of delivery to Japan" or "Register on the website."

The minimum number of users for measuring usability is 5. Jacob Nielsen, the founder of "Nielsen Norman Group," recommends running usability testing with 20 users.

Let's take a closer look at the most used usability metrics. We'll start with the **metrics for effectiveness measurement**.

Success score

However long your list of usability metrics is, the success score will probably be at the top of the list. Before we go into the details of usability, we have to find out if the design works. **Success**, or completion, means that a user managed to complete a task that they were given.

The basic formula for the success score is:

$$\text{Success Score} = \frac{\text{№ OF COMPLETED TASKS}}{\text{TOTAL № OF ATTEMPTS}}$$

The success score would be between 0 and 1 (or 0 and 100%). 0 and 1 are not just simple numbers. In this binary system, these numbers refer to the task being completed successfully or not. All the other specific situations are overlooked. Partial task success is considered a failure.

To have a more nuanced picture, UX researchers can **include tasks performed with errors in a separate group.** For example, the task is to purchase a pair of yellow shoes. The "partial success" options can be buying a pair of shoes of the wrong size, not being able to pay with a credit card, or entering the wrong data.

Let's say there were 20 users, 10 of whom successfully bought the right shoes, 5 chose the wrong type of delivery, 2 entered their address incorrectly, and 3 could not make the purchase. If we were counting just 0 or 1, we would have a rather low 50% success score. By counting all kinds of "partially successful" tasks, we get a whole spectrum.

Note! Avoid counting "wrong address" as 0,5 of success and adding it to the overall average, as it distorts the results.

Each "partially successful" group can tell us more than a general success score: using these groups, we can understand *where* the problem lies. We expect this more often from qualitative UX research, while quantitative gives us a precise but narrow-focused set of data.

For you to consider that a product has good usability, the success score doesn't have to be 100%. The average score is around 78%.

Now, let's move to the second of the most common usability testing metrics:

Number of errors

In user testing, an **error** is any wrong action performed while completing a task. There are two types of errors: slips and mistakes.

Slips are those errors that are made with the right goal (for example, a typo when entering the date of birth), and **mistakes** are errors made with the wrong goal (for instance, entering today's date instead of birth date).

There are two ways of measuring errors: measuring all of them (**error rate**) or focusing on one error (**error occurrence rate**).

To find the error occurrence rate, we have to calculate the total number of errors and divide it by the number of attempts. It is recommended to count *every* error, even the repetitive ones. For example, if a user tries to click an unclickable zone more than once, count each one.

$$\text{Error Rate} = \frac{\text{№ OF ERRORS}}{\text{TOTAL № OF ATTEMPTS}}$$

Error rate counts all possible errors. To calculate it, we need to define all possible slips and mistakes and the number of **error opportunities**. This number can be bigger or smaller depending on the complexity of the task. After that, we apply this simple formula:

$$\text{Error Occurrence Rate} = \frac{\text{TOTAL № OF ERRORS}}{\text{TOTAL № OF POSSIBLE ERRORS}}$$

Can there be a perfect user interface that prevents people from making typos? Unlikely. That is why the error rate seldom equals zero. Making mistakes is human nature, so having usability testing errors is fine.

As Jeff Sauro states in his "Practical Guide to Measuring Usability," only about 10% of the tasks are completed without any mistakes, and **the average number of errors per task is 0,7**.

Success score and error rate measure the effectiveness of the product. The following metrics are used to **measure efficiency**.

Task time

Good usability typically means that users can perform their tasks successfully and fast. The concept of task time metric is simple, yet there are some tricks to using it efficiently.

Task Time =

$$\frac{\text{TIME}_{\text{1ST USER}} + \text{TIME}_2 + \ldots + \text{TIME}_n}{\text{TOTAL № OF USERS}}$$

Having the average time, how do we know if the result is good or bad? There are some industry standards for other metrics, but there can't be any for task time.

Still, you can find an "ideal" task time. It's a result of an experienced user. To do this, you have to add up the average time for each little action, like "pointing with the mouse" and "clicking," using KLM (Keystroke Level Modeling). This system allows us to calculate this time quite precisely.

The task time metric is often measured to compare the results with older versions of the design or competitors.

Often, the difference in time will be tiny, but caring about time tasks is not just perfectionism. Remember, we live in a world where most people leave a website if it's not loading after 3 seconds. Saving those few seconds for users can greatly impact their user experience.

Efficiency

There are many ways of measuring efficiency. One of the most basic is time-based efficiency, which combines task time and success score.

Time-based Efficiency =

$$\frac{\sum_{j=1}^{R}\sum_{i=1}^{N}\frac{n_{ij}}{t_{ij}}}{NR}$$

R - number of users

N - number of tasks

n_{ij} - result of coming through scenario i by respondent j (1 or 0)

t_{ij} - time spent by respondent j to come through scenario i

Doesn't look basic, right? Not all formulas are easy to catch. It would take another article to explain this one in detail.

Tracking metrics is a whole science. If you want to dive deep into it, check out our list ofbest books about metrics (or leave it to theprofessional UX designers).

Now that we have figured out how to measure both effectiveness and efficiency, we get to **measuring satisfaction**, the key to user experience studies.

There are many satisfaction metrics, but we'll bring two that we consider the most efficient. For these metrics, the data is collected during usability testing by asking the users to fill in a questionnaire.

Single Ease Question (SEQ)

This is one of those easy and genius solutions that every UX researcher loves. Compared to all those complex formulas, this one is as simple as it gets: a single question is asked after the task.

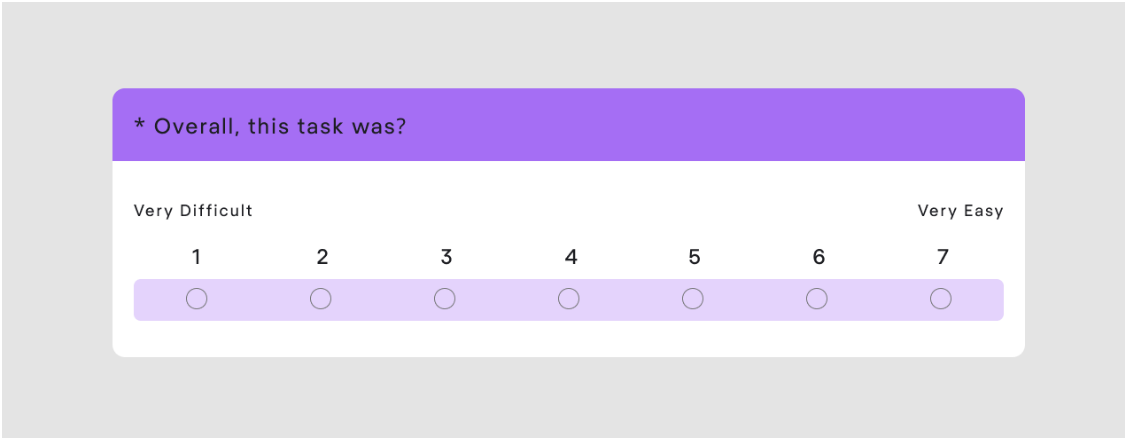

Image credit: measuringu.com

While most task-based usability metrics aim at finding objective parameters, SEQ is tapping into the essence of user experience: its subjectivity. Maybe the task took a user longer to complete, but they had no such impression.

What if the user just reacts slower? Or were they distracted for a bit? User's subjective evaluation of difficulty is no less important than the number of errors they made.

On average, users evaluate task difficulty at 4.8. Make sure your results are no less than that.

System Usability Scale (SUS)

For those who don't trust the single-question solution, there is a list of 10 questions known as the System Usability Scale. Based on the answers, the product gets a score on a scale from 0 to 100 (each question is worth 10 points).

System Usability Scale Questionnaire

	Strongly Disagree				Strongly Agree
1. I think that I would like to use this product frequently.	1	2	3	4	5
2. I found the product unnecessarily complex.	1	2	3	4	5
3. I thought the product was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this product.	1	2	3	4	5
5. I found the various functions in the product were well integrated.	1	2	3	4	5
5. I found the various functions in the product were well integrated.	1	2	3	4	5
6. I thought there was too much inconsistency in this product.	1	2	3	4	5
7. I imagine that most people would learn to use this product very quickly.	1	2	3	4	5
8. I found the product very awkward to use.	1	2	3	4	5
9. I felt very confident using the product.	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this product.	1	2	3	4	5

Image credit: Bentley university.

This scale comes in handy when you want to compare your product with the others: **the average SUS is 68 points**. Results over 80 are considered excellent.